

Statement of Work

1.1 Background : This new requirement is for the Research Service at the Central Arkansas Veterans Healthcare System (CAVHS) in Little Rock, Arkansas as part of the Shared Equipment Evaluation Program- Military Exposures Research Program (ShEEP-MERP), which is an Office of Research and Development (ORD) initiative that aims to:

- Improve our understanding of the effects of military exposures on Veteran health. Military exposures refer to contact with toxic agents, singly or in combination, acquired through ingestion, respiratory, or skin contact routes during military service (Deployment, Occupation, or Garrison).
- Advance the field of exposure assessment, which refers to qualitative and/or quantitative identification and measurement of toxic agent(s) to which a Veteran was exposed during military service. Exposure assessment may include the development of new methods and/or assays that will directly or indirectly identify and measure an exposure to a toxic agent using biological materials or surrogates.

Exposure to environmental chemicals is an important determinant of Veteran's Health. Our laboratory at CAVHS uses an approach called mass spectrometry to measure different classes of environmental chemicals in the blood of Veterans. We are making these measurements in Veterans and Military Personnel enrolled in large population health studies to understand occupational exposure to environmental chemicals. We are also developing simple self-administered blood collection and testing methods to study how were Veterans' lives and work impacts on their exposure to environmental chemicals. Currently our research focusses on per-fluorinated chemicals which we measure using an approach called liquid chromatography/mass spectrometry. However, we know that Veterans are also exposed to other classes of volatile/semi volatile, hydrophobic non-polar molecules environmental chemicals that include pesticides, fungicides, insecticides, flame retardants and halogenated chemicals that are not readily measured by this approach.

The Orbitrap mass analyzer instrumentation will enable us to measure these important classes of environmental chemicals in the blood and will ultimately support research to more broadly understand how the complement of environmental chemical exposures affects Veterans' Health. The Orbitrap mass analyzer instrument is both a major system and highly specialized. It employs a very unique function called ion trapping. Specifically, the orbitrap mass spectrometer we require is proprietary to the manufacturer and covered by multiple patents. This instrument will also be used to measure specific classes of lipids which are involved in cardiovascular and metabolic diseases and in weakening of the bones, all of which are of particular concern to the aging Veteran Population.

In June 2024, we submitted an application to the ShEEP-MERP to request funding specifically for this acquisition of the Orbitrap Exploris to conduct research related to the specific aims of the project for Exposomics. After review by a panel of scientific experts, the funds were approved by the Director of MERP, Rudolph Johnson, PhD., in November 2024 for this purchase.

There will not be any VA Sensitive information associated with this requirement.

1.2 Objective : Purchase a BRAND NAME ONLY – Thermo Scientific- Orbitrap Exploris Gas Chromatography-Mass Spectrometry (GC-MS) Quadrupole system with 60K + MSMS, including line items 1-23 listed below, for expsomics.

1.2.1 Line Items Required for Purchase include but not limited to:

Manufacturer Part Number: BRE7225543

Item #	Description/Part Number	Qty
1	Orbitrap Exploris GC with 60K + MSMS - BRE725543	1
2	TriPlus RSH SMART Standard liquid - 1R77010-2001	1
3	Merkur CountKit NA - BRE0017135	1
4	PERFORMANCE KIT - 1R120705-PERF	1
5	PWR CORD C13 TYPE B 125V MAX - 1R96000-98072	1
6	TRACE 1300/1600 PTV INJECTOR MODULE - 19070020	1
7	TRACE 1300/1600 SSL INJECTOR MODULE - 19070010	1
8	3 DAYS GC/GCMS ONSITE SUPPORT - 702-024403	1
9	PV PDU 2xL6-20R,2x6-20R,4x5-15R, NOT EU - C0202-99-00182	1
10	Powervar UPS SPII 6.0kVA, RoHS - C0012-01-00077	1
11	GC/GCMS Start-up Kit TRC1300/1600 - 19050720	1
12	RSH AUTOMATIC TOOL CHANGE STATION - 1R77010-1019	1
13	RSH HAND-HELD TERMINAL - 1R77010-1038	1
14	Wiley Registry 2023/NIST 2023 MS Library - B51002722	1
15	SW, COMPOUND DISCOVERER 3.3 SP3 - OPTON-31055	1
16	SW, TRACEFINDER 5.1 SP3 - OPTON-31001	1
17	OIL MIST FILTER w/NW25 ADAPTER/CLAMP - 1R76505-0036	1
18	GC/GCMS PTV ESSENTIALS KIT - 26096-1420PTV-BNDL	1
19	GC/GCMS SSL ESSENTIALS KIT - 26096-1420SSL-BNDL	1
20	Std Options Plan-Trace13x0/16x0 inj - 701-935400	2
21	Std Options Plan-TP RSH Tool Chng Statn - 701-058709	1
22	Ext War-TriPlus RSH SMART Std Lq - 701-059058	1
23	Extended Warranty with PM - Exploris GC MSMS - 701-058845	1

1.2.2 Salient characteristics include but not limited to:

- 1) Essential/significant physical, functional, or performance characteristics.

This is an integrated instrument system. Each of the components of the system have unique features and capabilities that are proprietary to the manufacturer and essential for government applications. Thermo Electron North America LLC sells mass spectrometers with Orbitrap mass analyzers and the required gas chromatograph quadrupole orbitrap system, which is highly

specialized and very unique. Alternative gas chromatograph mass spectrometer systems use different types of mass analyzers that do not have the sensitivity or ability to measure mass accurately at high resolution with high stability, which is offered by the Orbitrap mass analyzer.

The system contains technologies that are proprietary to the manufacturer.

The following is a non-exhaustive list of US patents covering the Orbitrap Exploris GC system:

- 1) US Patent No: 7,709,790 (Removable EI/CI Ion Source)
- 2) US Patent No: 7,714,283 (Improved Orbitrap Mass Analyzer Geometry)
- 3) US Patent No: 8,410,424 (Multiple Ion Injections)
- 4) US Patent No: 8,642,948 (Ion Cooling in C-Trap)
- 5) US Patent No: 8,853,620 (eFT Algorithm)
- 6) US Patent No: 10,748,755 (Orbitrap)

General Features

- Premium quantitative and qualitative performance with the fast-scanning High-Field Thermo Scientific Orbitrap mass analyzer
- Best-in-class resolving power, mass accuracy, sensitivity, and dynamic range
- Compatible Thermo Scientific Compound Discoverer software for chemometric analysis
- EI/CI Thermo Scientific™ ExtractaBrite™ ion source removable under vacuum through vacuum interlock
- Vent-free column exchange with source plug
- VeV low electron energy acquisition for increased molecular ion production

The Thermo Scientific Orbitrap Exploris GC 240 hybrid quadrupole Orbitrap mass spectrometer is a Thermo Scientific quadrupole-Orbitrap mass spectrometer based on proven hardware and instrument control software designs of the next-generation.

Thermo Scientific mass spectrometers and incorporates extensive customer and service engineer feedback.

Hardware features

Ion source

- Thermo Scientific™ ExtractaBrite™ Electron Ionization (EI) source
- Ion source includes ion volume, repeller, source lenses, RF lens and dual filaments in all ionization modes, programmable from 50 °C to 350 °C
- VeV tuning allows optimized low electron energy acquisition down to 8 eV
- Chemical Ionization (CI) source for acquisition with Positive Ion Chemical Ionization (PCI) and Negative Ion Chemical Ionization (NCI)
- Remove entire ion source or change to CI source in under 2 minutes without venting
- Vent-free column exchange with patented source plug

Combination EI/PCI/NCI ion volume can be used without need for source interchange.

Ion optics

Advanced active beam guide (AABG)

Axial field reduces noise by preventing neutrals and high- velocity clusters from entering the quadrupole mass filter using double bent design geometry.

Advanced quadrupole technology (AQT)

- Segmented quadrupole mass filter for precursor ion selection with variable precursor isolation width from 0.4 to 1,200 Da

- MS/MS and SIM precursor ion selection with high transmission from m/z 30 to 2,000

Ion-routing multipole (IRM)

- Robust ion trapping for higher energy collisional dissociation (HCD)
- Nitrogen collision gas

Automatic gain control (AGC)

Reliable AGC measurements for controlled injection of the number of ions.

Thermo Scientific Orbitrap mass analyzer

- High-Field Orbitrap mass analyzer
- Low noise detection pre-amplifier
- 4 kV central electrode voltage

Vacuum system

- A compact single turbo pump design providing the adequate vacuum in five stages for the aluminum high- vacuum analyzer chambers
- Advanced vacuum technology reduces pressure in the ultra-high vacuum regions, enhancing transmission of ions to the Orbitrap mass analyzer

Optional hardware

Direct sample probe system option

- Switch to probe <3 min with GC undisturbed
- Available in two styles: rapid heating filament Direct- Exposure Probe (DEP, capable of flash vaporization or pyrolysis at up to 1,600 °C) or slower volatilization Direct-Insertion Probe (DIP, capable of accommodating powders and solid samples in a quartz or aluminum crucible) up to 450 °C

High-Field Orbitrap Mass Analyzer

Performance specifications

Mass range m/z 30–3,000

Orbitrap mass analyzer resolution Up to 240,000 at m/z 200

Scan rate* Up to 40 Hz at resolution setting 7,500 at m/z 200

Mass accuracy* External calibration achieves <3 ppm RMS drift over 24 hours

Internal lock mass calibration achieves <1 ppm RMS drift over 24 hours

Data acquisition system

Data system

- High-performance PC with Intel® microprocessor
- High-resolution LED color monitor

- Microsoft® Windows® 10 Enterprise (Long Term Service version) operating system
- High-speed real-time data acquisition and instrument control
- Automatic calibration of all ion transfer and analysis parameters via instrument control software

Thermo Scientific Orbitrap Exploris instrument control software

- Tune application for instrument mass and system calibrations and checks, diagnostics, and manual data acquisition
- Method Editor with a comprehensive application- specific template library, method setup supported by tooltips, and a drag-and-drop user interface to facilitate method development
- Consistent instrument control software whether using Thermo Scientific™ Xcalibur™ or Thermo Scientific™ Chromeleon™ Chromatography Data Systems (CDS) for data acquisition

Included software

Thermo Scientific Xcalibur software

- Xcalibur software is the control software for the next- generation Thermo Scientific mass spectrometer portfolio
- Accelerates familiarization and reduces training needs

Operation modes

Resolution settings

Ranging from 7,500 to 240,000 at m/z 200

Scan functions

Full scan

High sensitivity, high selectivity full scan for targeted and untargeted analyses.

tSIM

- Targeted SIM with Mass List Table
- With Targeted Mass Filter for ddMS2
- Isolation Width, Resolution, Polarity, Microscans set values are definable compound-dependent (w/o msx)
- Isolation Width: 0.4 u to 50 u
- Multiplexing for up to 20 compounds
- MSX ID, multiplexing groups definable
- Isolation Width set values are definable compound- dependent (w/ msx)

tMS2

- Targeted MS2 with Mass List Table
- Isolation Width, HCD Collision Energy, Polarity, Microscans set values are definable compound- dependent (w/o msx)
- Isolation Width: 0.4 u to 50 u
- Multiplexing for up to 20 compounds
- Isolation Width set values can be defined compound- dependent (w/ msx)

SIM by Data-Dependent Acquisition (DDA) following a master scan (i.e., a Full MS scan or tSIM scan):

- With up to Top 100 for ddSIM
- With Targeted Mass Filter
- 'Number of Scans' (TopN) and 'Cycle Time' (Top Speed) option
- Isolation Width: 0.4 u to 50 u
- By performing a dependent scan on the most intense ion, if no target mass is found (optional)

MS2 by Data-Dependent Acquisition (DDA) following a master scan (i.e., a Full MS scan or a tSIM scan):

- With up to Top 100 ddMS2
- With Targeted Mass Filter
- 'Number of Scans' (TopN) and 'Cycle Time' (Top Speed) option
- Isolation Width: 0.4 u to 50 u
- HCD Collision Energy set value is definable per compound
- By performing a dependent scan on the most intense ion, if no target mass is found (optional)

General

- Multiple experiments can be set up within one method
- One experiment can contain combinations of scans
- Collision Energy Mode' can be selected: 'Fixed' and 'Stepped'

Filters

Filters guide data-dependent (discovery and conformational) decisions on the fly and in real time. To achieve optimum results when applying application- and sample-dependent filter settings, the user is guided with appropriate application-dependent parameter settings and tool tips with tailored recommendations and detailed 'learn more' sections.

Filters can be selected as follows:

- Dynamic Exclusion
- Intensity
- Precursor Fit
- Targeted Inclusion
- Targeted Exclusion
- Apex Detection
- Precursor Selection Range

- Compatibility** 100–240 VAC, 3.5 A, 50/60Hz
- Dimensions** – Dimensions (w, d, h) • 954 × 1,036 × 703 mm (38 × 41 × 28 in)
- Weight**- Weight • 156 kg (344 lb) including GC and one injector, without data system, vacuum rough pumps, and optional items.
- Industry Standard** – The system must meet or surpass the highest performance standards for commercially available instrument of this type and configuration. This includes the specifications detailed in the table below:

Performance specifications	
Mass range	<i>m/z</i> 30–3,000
Orbitrap mass analyzer resolution	Up to 240,000 at <i>m/z</i> 200
Scan rate*	Up to 40 Hz at resolution setting 7,500 at <i>m/z</i> 200
Mass accuracy*	External calibration achieves <3 ppm RMS drift over 24 hours Internal lock mass calibration achieves <1 ppm RMS drift over 24 hours
Sensitivity	EI: 100 fg octafluoronaphthalene on column, scanning <i>m/z</i> 50–300, S/N 10,000:1 EI IDL**: 6 fg octafluoronaphthalene derived at the 99% confidence level PCI: 10 pg benzophenone on column, scanning <i>m/z</i> 80–230, S/N 150:1
Dynamic range	>10 ⁶ analytical dynamic range* >5,000 within a single Orbitrap mass analyzer spectrum
Polarity switching	One Full Scan cycle*** <700 ms equals >1.4 Hz One tSIM Scan cycle*** <600 ms equals >1.6 Hz
Multiplexing	Up to 20 precursors per scan

- e) **Purpose – Research** – The system is required for analysis of environmental chemicals in blood samples collected from veterans and military personnel to support studies of the health effects of exposure to these chemicals. These studies are supported by VA, the Department of Defense, the National Institutes of Health and involve studies of subjects enrolled in large population health cohorts. Funding for this instrument system comes from the VA Military Exposures Research Program Shared Equipment Evaluation Program. The application process involved scientific peer review which ratifies the suitability of the instrument system for our needs.